Feasible Range of the Biomarker Moderation Parameter Under AR(1) and Compound-Symmetry Correlation Structures

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Purpose. This whitepaper determines how the choice of temporal correlation structure in the covariance-moderation data-generating process (DGP) affects the largest biomarker moderation parameter c_{bm} for which the joint correlation matrix remains positive definite. It replaces a grid search with an exact closed-form ceiling, evaluates that ceiling on the three trial designs of paper 01 (`analysis/report/01-dgp-mean-moderation-vs-mvn/`), and supplies the numbers requested by the `[TO RE-VERIFY]` note at the end of that paper's Appendix B.7.

It then asks a design question: which features of a trial schedule (number of visits, spacing between visits, the gap at a treatment switch, the number of on/off switches, the share of visits on drug) determine how large a biomarker-response correlation a simulation can test? The ceiling is the largest testable c_{bm}: a value above it cannot be simulated, because no valid covariance matrix carries it. The figures in Section 5 are intended as design aids for that decision. Throughout, "largest testable c_{bm}" and "ceiling" mean the same thing.

Review note (2026-10-07). Every computation here uses the paper 01 production value $\rho = 0.7$. The published code at `58b32a9` uses $\rho = 0.8$ for all three components (inspected, `58b32a9` `vignettes/Produce_Publication_Results_1_generate_data.Rmd`, lines 317 to 341). At $\rho = 0.8$ the published construct's ceiling without carryover is 0.256 in Hybrid and CO and 0.262 in OL+BDC, and 0.893 at any positive half-life in Hybrid and OL+BDC (`docs/36`, Section 4.2; recomputed in this review). The "published specification" rows below are therefore that specification at the paper 01 production parameters, not at the published ones. Two further results postdate this paper: the separable AR(1) form of `docs/35`, under which the ceiling is 0.440 at $\rho = 0.8$, and 0.474 (Hybrid 0.480) at $\rho = 0.7$; and the finding of `docs/37` that the published step coupling makes the on/off coupling gap zero, so that the interaction vanishes at any positive half-life, which graded coupling removes. Paper 01 has also been restructured since this paper was written; its Appendix B.7 and the `[TO RE-VERIFY]` note cited below no longer exist, and its current ceiling table (Section "Room for larger effect sizes") reports these values.

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Notation index and glossary

Notation follows `analysis/report/NOTATION.md`. Symbols marked *local* are defined here and are not part of the canonical set.

Table 1: Notation index

Symbol	Meaning	Status
B, BL	biomarker and baseline level	canonical
TV, PB, BR	the three response components	canonical
c_{bm}	covariance-moderation correlation: on-drug correlation of B with BR	canonical
c_{bm}^*	ceiling: largest c_{bm} with a positive definite matrix	local
R	joint correlation matrix of $(B, \mathrm{BL}, TV, PB, BR)$	local
M	correlation block of the three response factors	local
A, C	within-factor and cross-factor blocks of M	local
$\tilde{r}$	biomarker coupling vector, nonzero on the BR positions	local
v	coupling pattern, $\tilde{r} = c_{bm} v$	local
ϕ_t	residual fraction at off-drug occasion t, $(1/2)^{t_{sd,t}/t_{1/2}}$	local
n	number of measurement occasions on a path	local
n_{on}	number of occasions with the coupling on (CS closed form)	local
t, s	occasion indices (s local)	canonical
w_t	cumulative calendar week of occasion t	local
d_t, d	gap between occasions $t-1$ and t; a gap at a switch	local
ρ	within-factor correlation (AR(1) or CS)	canonical
c_1	same-occasion cross-factor correlation (code `c.cf1t`)	local
$c_\times$	cross-factor correlation at distinct occasions (code `c.cfct`)	local
K_1, K_2	3×3 coefficient matrices of I_n and J_n in the CS form of M	local
I_n, J_n	identity and all-ones matrices of order n	local
a, a_n	constants of the CS closed form, Section 5.1	local
u	a generic vector in the AR(1) precision identity, Section 5.5	local
t_{sd}, $t_{sd,t}$	time since discontinuation (at occasion t)	canonical
$t_{1/2}$	carryover half-life, weeks	canonical
λ_w, k	Weibull rate and shape of the decay family	canonical

Glossary. *`covar`*: the package's covariance-moderation process (`dgp_architecture = 'mvn'`, formerly Architecture B). *`orig`*: the published Hendrickson et al. [1] implementation at commit `58b32a9`. *Compound symmetry (CS)*: one constant within-factor correlation for every pair of occasions. *Graded coupling*:

$v_t = 1$ on drug, ϕ_t off drug after exposure, 0 before first exposure. *Step coupling*: $v_t = 1$ wherever any residual drug effect remains. *Binding path*: the path with the smallest ceiling in a design. *Positive-definiteness repair*: `corpcor::make.positive.definite()` applied when the covariance matrix fails the check.

1. Summary

The ceiling on c_{bm} is available in closed form, $c^*_{bm} = (v^\top M^{-1} v)^{-1/2}$, where M is the correlation block of the three response factors and v is the on-drug and off-drug coupling pattern with c_{bm} factored out. We computed it for every path of the CO, Hybrid, and OL+BDC designs under a crossed set of correlation structures, and on synthetic schedules that vary one design feature at a time, and under alternative shapes of carryover decay. Nine findings follow.

1. **The reference value sits at the ceiling.** Under the covariance-moderation DGP as implemented (Architecture B, `dgp_architecture = 'mvn'`), the ceiling at $t_{1/2} = 0$ is 0.456 in Hybrid and 0.452 in OL+BDC. Paper 01's reference value $c_{bm} = 0.45$ therefore lies within 0.002 to 0.006 of the boundary in its no-carryover cells.
2. **AR(1) widens the feasible range at the production value of ρ.** At $\rho = 0.7$, replacing compound symmetry (CS) with AR(1) in the within-factor block raises the ceiling in every design, by 0.09 to 0.25 with a constant cross-factor block and by 0.21 to 0.45 with a decaying one.
3. **The widening is not general.** At $\rho = 0.3$, compound symmetry admits a higher ceiling than AR(1) in all three designs, and at $\rho = 0.5$ it still does in OL+BDC.
4. **The ceiling depends on carryover.** Under Architecture B the ceiling rises with the carryover half-life, to 0.544 (Hybrid) and 0.533 (OL+BDC) at $t_{1/2} = 1$.
5. **The published configuration cannot run the reference value without carryover.** For the specification described as Hendrickson et al.'s published code (CS, constant cross-factor block, step coupling), the ceiling at $t_{1/2} = 0$ is 0.346 to 0.352 at $\rho = 0.7$. It jumps to 0.841 at any positive half-life in Hybrid and OL+BDC, but not in CO. At the published $\rho = 0.8$ the corresponding values are 0.256 to 0.262 and 0.893, so even the published $c_{bm} = 0.3$ is infeasible without carryover (`docs/36`, Section 4.2; added 2026-10-07).
6. **Values just above the ceiling are altered without warning.** The positive-definiteness repair applied by `buildSigma()` perturbs the specified correlations by at most 0.004 just above the ceiling, and reports nothing unless `verbose = TRUE`.
7. **Under compound symmetry, schedule design is irrelevant to the ceiling.** It depends only on the number of visits n, the number of visits with the coupling on n_{on}, and the correlation parameters, through an

exact closed form. Spacing, the gap at a switch, the number of switches, and the order of on-drug visits have no effect.

8. **Under AR(1), schedule design governs the ceiling.** Each additional on/off switch lowers it, sharply when visits are close together; with 16 weekly visits it falls from 0.51 with one switch to 0.15 with fifteen. A short gap at the switch lowers it (0.33 at a half-week gap, 0.72 at an eight-week gap, at fixed study length). Uneven spacing in the paper 01 Hybrid and OL+BDC schedules costs 0.04 to 0.09 relative to even spacing over the same span.
9. **The shape of carryover decay would matter, moderately.** If the biomarker-response correlation decayed as one of the Weibull shapes of paper 02 rather than exponentially, faster washout would lower the ceiling toward its no-carryover value and a heavier tail would raise it, by up to 0.09 in the paper 01 designs and 0.13 in a multi-cycle design, at short half-lives. The package currently decays the correlation exponentially only, so this is a counterfactual.

2. Evidence basis

Evidence labels follow the project convention: **verified** (computed and checked in this analysis), **inspected** (confirmed by reading code or documents), **inferred** (follows from verified results but not checked directly), and **unverified**.

- The matrix builder used here reproduces the package's `buildSigma()` correlation matrix for Architecture B to 1.1×10^{-16} on all eight design paths at all three half-lives. (Verified.)
- The closed-form ceiling agrees with a bisection on the smallest eigenvalue of the full matrix to 2.0×10^{-11} across all 192 path configurations. (Verified.)
- The compound-symmetry and step-coupling levels implement the specification of Hendrickson et al.'s published code at commit `58b32a9` as described in paper 01 Sections 2.2.4 and B.2 to B.4. That commit is not present in this repository; the only vendored copy (`analysis/scripts/quick-sim/hendrickson-original-comparison/`) is the 2024 revision. Ceilings for those levels are therefore ceilings for the specification as described, not for the published code. (Unverified against the source.) *Update 2026-10-07:* a copy of `58b32a9` is now held at `analysis/data/quick-sim/hendrickson-58b32a9/`. Its `R/generateData.R` matches the specification used here: CS within factors, a constant cross-factor block, and step coupling on a nonzero carryover-adjusted response mean at every occasion, including the first (inspected, lines 83 to 138). `docs/36` ran that code and reproduced the ceilings (verified there). The published parameters differ, however: $\rho = 0.8$, not 0.7 (review note above).

- The ceiling does not depend on how carryover is applied to the response mean, and in particular not on the recursive-versus-anchored error in that calculation corrected in commit `955c604`. Under Architecture B the coupling is set from t_{sd} and λ directly (`R/generateData.R:341-356`), and the response means do not enter the correlation matrix (inspected). Under the step coupling only whether the response mean is nonzero matters, and the recursive and anchored forms have the same zero pattern (inferred from the two formulas). The error does affect the paper 01 power results; that is addressed in `docs/33-paper01-dgp-architecture-review.md`.
- The run used the host R session without `renv`, at commit `c0b31f5` with uncommitted working-tree changes. No function in the package source was modified.

3. Method

3.1 The correlation matrix

For one participant on one randomization path with n measurement occasions at cumulative weeks $w_1 < \cdots < w_n$, the DGP draws $(B, \mathrm{BL}, TV_{1:n}, PB_{1:n}, BR_{1:n})$ from a multivariate normal distribution. On the correlation scale the matrix is

$$R = \begin{pmatrix} 1 & 0 & \tilde{r}^\top \\ 0 & 1 & 0^\top \\ \tilde{r} & 0 & M \end{pmatrix}, \qquad M = \begin{pmatrix} A & C & C \\ C & A & C \\ C & C & A \end{pmatrix},$$

where A is the $n \times n$ within-factor block, C the cross-factor block, and $\tilde{r}$ the biomarker coupling vector, nonzero only on the BR positions. The baseline row factors out of every determinant and is dropped below.

3.2 The exact ceiling

Write $\tilde{r} = c_{bm} v$. By the Schur complement on the biomarker row, R is positive definite if and only if M is positive definite and $1 - c_{bm}^2 \, v^\top M^{-1} v > 0$. The quantity $c_{bm}^2 \, v^\top M^{-1} v$ is the coefficient of determination from regressing the biomarker on all response variables, so the condition states only that the biomarker cannot be more than perfectly predictable. The ceiling is

$$c_{bm}^* = \left(v^\top M^{-1} v \right)^{-1/2},$$

and it is attained: at $c_{bm} = c_{bm}^*$ the smallest eigenvalue of R is exactly zero. A design's feasible range is the minimum of c_{bm}^* over its paths, since every path's matrix must be valid at the common c_{bm}.

3.3 Crossed factors

The two blocks that "AR(1)" touches in the implementation, and the coupling rule that sets v, were crossed so that the effect of each could be read with the others held fixed.

Table 2: Crossed factors and their levels in the ceiling computation

Factor	Levels
Within-factor block A	CS: $A_{ts} = \rho$ for $t \neq s$. AR(1): $A_{ts} = \rho^{\lvert w_t - w_s \rvert}$
Cross-factor off-diagonal C_{ts}, $t \neq s$	Constant: $c_\times$. Decaying: $c_\times \rho^{\lvert w_t - w_s \rvert}$
Coupling v at off-drug occasions	Graded: ϕ_t. Step: $\mathbf{1}[\phi_t > 0]$
Carryover half-life $t_{1/2}$	0, 0.5, 1.0 weeks
Design and path	CO (2 paths), Hybrid (4), OL+BDC (2)

In every configuration, $C_{tt} = c_1$ on the diagonal and $v_t = 1$ at on-drug occasions. The residual fraction is $\phi_t = (1/2)^{t_{sd,t}/t_{1/2}}$ at an off-drug occasion that follows drug exposure when $t_{1/2} > 0$, and $\phi_t = 0$ otherwise, including at off-drug occasions before first exposure.

Two corners of the first three rows correspond to named processes.

- **Architecture B (`covar`)**: AR(1), decaying cross-factor, graded. This is what `buildSigma()` constructs for `dgp_architecture = 'mvn'`.
- **Published specification (`orig`)**: CS, constant cross-factor, step.

The remaining configurations are counterfactual combinations used to isolate one choice at a time.

3.4 Parameters and designs

Production parameters were taken from the paper 01 driver (`analysis/scripts/quick-sim/01-dgp-prototyp` lines 181-182): $\rho = 0.7$ for all three response factors, $c_1 = 0.2$, $c_\times = 0.1$. The designs are those of the same driver (lines 65-103).

Table 3: Occasion schedules and paths of the CO, Hybrid and OL+BDC designs

Design	Occasions (weeks)	Paths
CO	2.5, 5, …, 20	A: on drug at occasions 1-4. B: on drug at occasions 5-8

Design	Occasions (weeks)	Paths
Hybrid	4, 8, 9, 10, 11, 12, 16, 20	A-D: open label, blinded discontinuation, one crossover cycle
OL+BDC	4, 8, 12, 16, 17, 18, 19, 20	A: off drug at 19-20. B: off drug at 18-20

Because the three response factors share one value of ρ, the loop-order dependence of the cross-factor decay rate noted in paper 01 Appendix B.5.2 does not arise.

A sensitivity sweep over $\rho \in \{0.3, 0.5, 0.7, 0.9\}$ was run for Architecture B and for CS with a constant cross-factor block under both couplings.

4. Results

4.1 Ceilings at production parameters

Design-level ceilings (minimum over paths), reported as $t_{1/2} = 0 \,/\, 0.5 \,/\, 1$ (verified):

Table 4: Design-level c_{bm} ceilings by correlation structure, coupling and half-life

A	C off-diagonal	Coupling	CO	Hybrid	OL+BDC
AR(1)	decaying	graded (`covar`)	.619 / .619 / .619	.456 / .506 / .544	.452 / .500 / .533
AR(1)	decaying	step	.619 / .494 / .494	.456 / .519 / .519	.452 / .519 / .519
AR(1)	constant	graded	.591 / .591 / .591	.452 / .490 / .514	.438 / .472 / .491
AR(1)	constant	step	.591 / .430 / .430	.452 / .462 / .462	.438 / .463 / .463
CS	constant	graded	.346 / .346 / .346	.346 / .370 / .411	.352 / .386 / .456
CS	constant	step (`orig`)	.346 / .346 / .346	.346 / .841 / .841	.352 / .841 / .841
CS	decaying	graded	.164 / .164 / .164	.247 / .265 / .299	.151 / .170 / .214
CS	decaying	step	.164 / .164 / .164	.247 / .845 / .845	.151 / .832 / .832

The binding path for Architecture B is path A in Hybrid and OL+BDC at

$t_{1/2} = 0$, switching to path B in Hybrid at $t_{1/2} = 1$. In CO it is path A at $t_{1/2} = 0$ and path B at positive half-lives.

Figure 1 shows the two named processes. Values above a curve cannot be simulated under that structure.

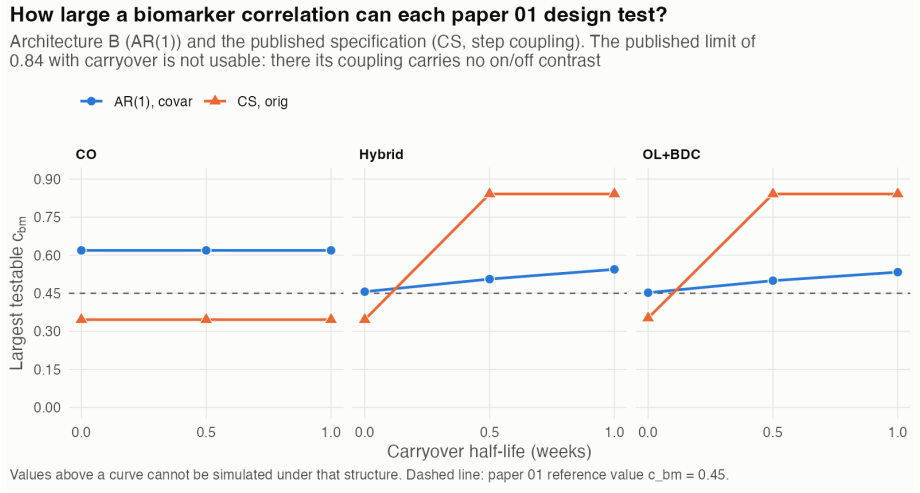


Figure 1: Largest testable c_{bm} for each paper 01 design, by carryover half-life, under Architecture B and the published specification, at $\rho = 0.7$. The published limit of 0.84 at positive half-lives is not usable: in that regime its coupling carries no on-drug versus off-drug contrast (Section 4.6).

4.2 The response block on its own

The response block M must itself be positive definite for any $c_{bm} > 0$ to be feasible. Its smallest eigenvalue at production parameters is (verified):

Table 5: Smallest eigenvalue of the response block M by structure and design

A	C off-diagonal	CO	Hybrid	OL+BDC
AR(1)	decaying	0.288	0.074	0.074
AR(1)	constant	0.331	0.093	0.093
CS	constant	0.200	0.200	0.200
CS	decaying	0.034	0.039	0.015

For CS with a constant cross-factor block, the analytical expression $(1 - \rho) - (c_1 - c_\times)$ given in paper 01 Appendix B.7 predicts $0.3 - 0.1 = 0.2$, which matches exactly. The design-dependent values for AR(1) reflect the unequal spacing of the Hybrid and OL+BDC schedules, whose one-week gaps produce strongly correlated adjacent occasions.

4.3 The effect of AR(1), isolated

Holding the cross-factor block and the coupling fixed (graded coupling, $t_{1/2} = 0$), replacing CS by AR(1) in the within-factor block changes the ceiling as follows (verified):

Table 6: Ceiling change from CS to AR(1) by cross-factor block and design

Cross-factor block	CO	Hybrid	OL+BDC
Constant	0.346 to 0.591	0.346 to 0.452	0.352 to 0.438
Decaying	0.164 to 0.619	0.247 to 0.456	0.151 to 0.452

At the production value of ρ, AR(1) widens the feasible range under either cross-factor choice. The size of the gain depends on the cross-factor block, because the two choices interact. A decaying cross-factor block combined with a CS within-factor block produces the lowest ceilings in the table, consistent with paper 01 Appendix B.7's statement that this combination lowers the feasible range substantially. That combination asserts that a factor's correlation with *another* factor decays with lag while its correlation with *itself* does not, and M is then close to singular (Section 4.2).

4.4 Dependence on ρ

Design-level ceilings at $t_{1/2} = 0$, reported as CO / Hybrid / OL+BDC (verified; "not PD" means M itself is not positive definite):

Table 7: Design-level ceilings at $t_{1/2} = 0$ across values of ρ

ρ	CS, constant cross, graded	AR(1), decaying cross, graded
0.3	.520 / .504 / .504	.502 / .475 / .428
0.5	.458 / .458 / .458	.549 / .489 / .448
0.7	.346 / .346 / .352	.619 / .456 / .452
0.8 (published)	.256 / .256 / .262	.627 / .291 / .303
0.9	not PD	.449 / not PD / not PD

The $\rho = 0.8$ row was added on 2026-10-07 (verified by an independent closed-form computation in that review; it agrees with `docs/36` and with paper 01's ceiling table). It is not in `rho.csv` or Figure 2, which cover the four values of the original sweep.

The two structures respond to ρ differently. Under CS the ceiling falls steadily as ρ rises, and M reaches singularity at $\rho = 0.9$, where $(1 - \rho) - (c_1 - c_\times) = 0$. Under AR(1) the ceiling rises with ρ through 0.7 in CO and OL+BDC, and in

Hybrid peaks at $\rho = 0.5$ before falling. At $\rho = 0.9$ it collapses: the one-week gaps in Hybrid and OL+BDC make adjacent occasions so strongly correlated that M is no longer positive definite, and the CO ceiling falls to 0.449. The added $\rho = 0.8$ row shows the fall in Hybrid and OL+BDC beginning before that point, to 0.291 and 0.303, while CO is still rising (0.627).

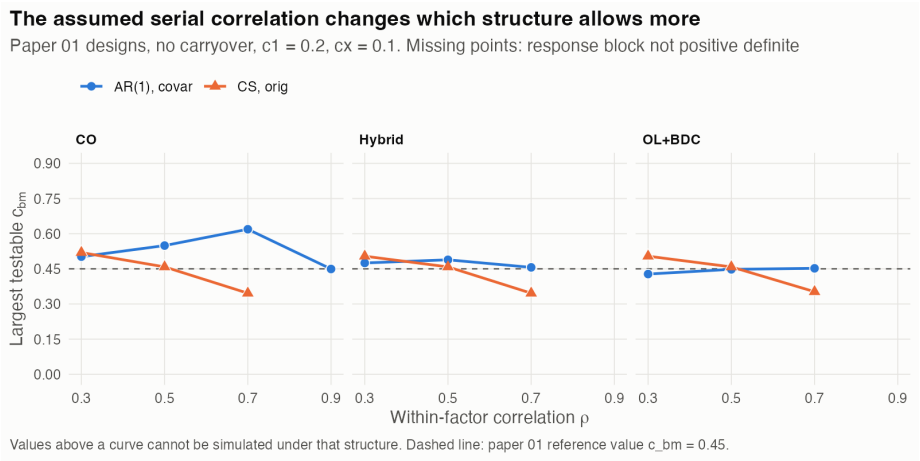


Figure 2: Largest testable c_{bm} by the assumed within-factor correlation ρ, for the paper 01 designs without carryover. Missing points mark values of ρ at which the response block is not positive definite.

The consequence is that "AR(1) expands the feasible range of c_{bm}" is true at $\rho = 0.7$ and false at $\rho = 0.3$. At $\rho = 0.5$ it holds in CO and Hybrid but not in OL+BDC.

4.5 Dependence on carryover

Under Architecture B the ceiling rises with carryover in Hybrid (0.456 to 0.544) and OL+BDC (0.452 to 0.533), and is flat in CO (0.619). The following interpretation is inferred and consistent with Appendix B.7 of paper 01: graded off-drug entries make v smoother across occasions, moving it toward the constant direction in which M is best conditioned. CO is flat because its binding path at positive half-lives is the placebo-first path B, whose coupling pattern does not change with carryover, since $\phi_t = 0$ before first exposure.

The practical implication is that a c_{bm} grid chosen to be feasible at positive half-lives may be infeasible at $t_{1/2} = 0$. The no-carryover cell is the binding constraint.

4.6 The ceiling reversal under step coupling

Paper 01 Appendix B.7 states that the published coupling has the lower ceiling at $t_{1/2} = 0$ and the higher ceiling at any positive half-life. For Hybrid and OL+BDC this is confirmed: 0.346 against 0.456 at $t_{1/2} = 0$, and 0.841 against 0.50 to 0.54 at positive half-lives (verified, $\rho = 0.7$). The mechanism is the one given there. Once any carryover is present, the step coupling assigns the full c_{bm} at every post-exposure occasion, so v is constant on paths that begin on drug, and a constant vector aligns with the leading eigenvector of a compound-symmetric block.

Two qualifications are needed.

- **The reversal does not occur in CO.** There the published ceiling is 0.346 at every half-life. CO's placebo-first path has pre-exposure zeros in v at every half-life, so v never becomes constant. Appendix B.7 states the reversal without this qualification.
- **The jump depends on compound symmetry.** With step coupling but an AR(1) within-factor block, the ceiling at positive half-lives is 0.519 rather than 0.841. The large apparent gain in the published configuration therefore requires both the step and the compound-symmetric block.

As Appendix B.7 argues, the higher ceiling is not an advantage: it is available precisely because the coupling vector has stopped carrying any on-drug versus off-drug contrast.

4.7 Behavior of the positive-definiteness repair

The simulation driver calls `generateData()` with `makePositiveDefinite = TRUE`. When the covariance matrix fails `is.positive.definite()`, `buildSigma()` replaces it with `make.positive.definite(sigma, tol = 1e-3)` (`R/generateData.R:377-388`) and issues a warning only when `verbose = TRUE` (inspected). A second repair follows it: if the Cholesky factorization fails, the sampler factors `make.positive.definite(sigma, tol = 1e-3)` instead, with no warning at all (`R/generateData.R:391-393`, inspected 2026-10-07; see also `docs/36`, Section 6.7).

On each design's binding path for Architecture B at $t_{1/2} = 0$, the specified correlations were compared with those in the repaired matrix at $0.99\,c^*_{bm}$ and $1.01\,c^*_{bm}$ (verified):

Table 8: Largest change in correlations after repair near the ceiling, by design

Design	c_{bm}	Largest change in r
CO	0.6127	0.0000

Design	c_{bm}	Largest change in r
CO	0.6251	0.0041
Hybrid	0.4516	0.0000
Hybrid	0.4608	0.0008
OL+BDC	0.4477	0.0000
OL+BDC	0.4567	0.0008

Below the ceiling the repair does not act, so the paper 01 production run at $c_{bm} = 0.45$ drew from the matrices as specified. This is consistent with an earlier check showing a smallest eigenvalue of 0.0021 in OL+BDC and 0.0057 in Hybrid at that value. Just above the ceiling the repair acts, and the perturbation is small enough to pass unnoticed. A run at $c_{bm} = 0.46$ in OL+BDC would therefore be silently perturbed rather than rejected.

5. Which design decisions change the largest testable c_{bm}

Section 4 evaluated three fixed designs. This section varies the schedule itself, one feature at a time, to identify which design decisions move the largest testable c_{bm} and by how much. Each correlation structure is paired with its own cross-factor form: AR(1) with decaying cross-factor entries (Architecture B, `covar`) and CS with constant ones (the published specification, `orig`). All schedules use the production parameters and no carryover, where the graded and step couplings coincide, so that only the correlation structure and the schedule vary.

5.1 Compound symmetry: schedule design is irrelevant

Under CS every block of M is a linear combination of I_n and the all-ones matrix J_n, so $M = K_1 \otimes I_n + K_2 \otimes J_n$ with

$$K_1 = (1-\rho)I_3 + (c_1 - c_\times)(J_3 - I_3), \qquad K_2 = \rho I_3 + c_\times(J_3 - I_3).$$

M^{-1} then separates into a part acting on the constant direction and a part acting on its complement, and for a 0/1 coupling pattern with n_{on} of n visits on,

$$\frac{1}{c_{bm}^{*\,2}} = a\,\frac{n_{\text{on}}(n - n_{\text{on}})}{n} + a_n\,\frac{n_{\text{on}}^2}{n}, \qquad a = [K_1^{-1}]_{33}, \quad a_n = [(K_1 + nK_2)^{-1}]_{33}.$$

This expression agrees with the numerical ceiling to 2.8×10^{-16} in every schedule examined (verified). It contains only n, n_{on}, ρ, c_1 and $c_\times$. The times of the visits, the gaps between them, the number of switches, and the order of on-drug

and off-drug visits do not appear. The reason is structural: under CS every pair of visits is equally correlated, so the matrix cannot distinguish one visit from another, and only how many visits carry the coupling can matter.

Figure 3 shows this directly. It collects every schedule in the analysis with 8 visits of which 4 are on drug. The CS limit is 0.346 in all twenty, while the AR(1) limit ranges from 0.33 to 0.74 across the same schedules.

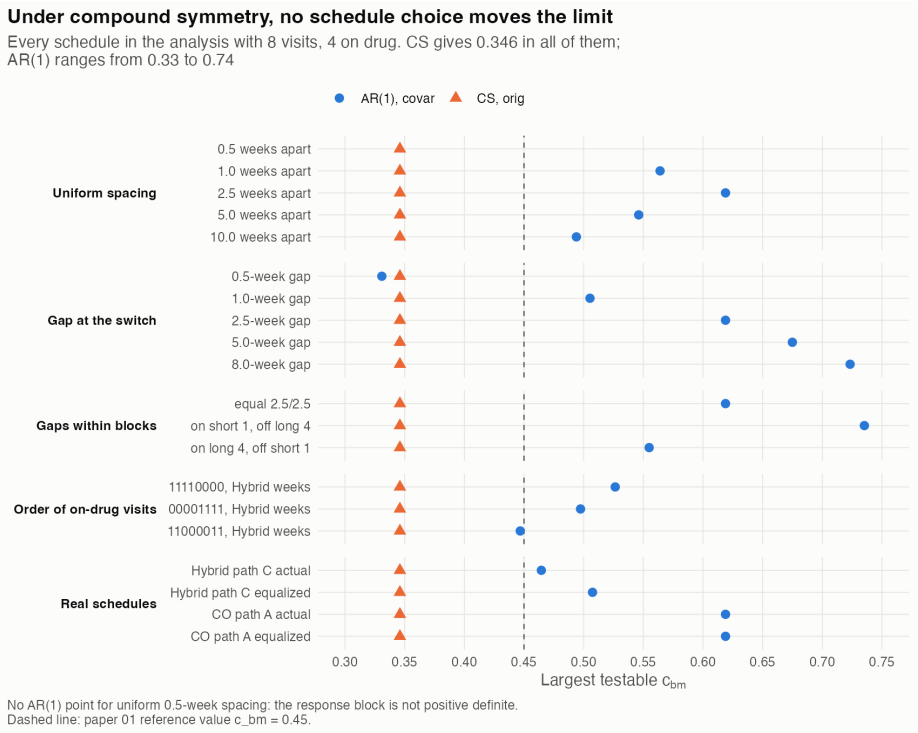


Figure 3: Largest testable c_{bm} for every 8-visit, 4-on-drug schedule in the analysis, grouped by the design feature varied. The compound-symmetry limit is identical in every schedule.

What does matter under CS is the number of visits and the share with the coupling on, and Figure 4 is therefore a complete lookup for any CS design at these parameters. The limit falls roughly as $n^{-1/2}$ at a fixed share, and is lowest when about half the visits carry the coupling. At these parameters no CS design with 16 or more visits can test $c_{bm} = 0.45$; with 8 visits only the most unbalanced shares (1 or 7 of 8) can.

The cross-factor parameters enter almost entirely through their difference $c_1 - c_\times$ (verified, Hybrid path C):

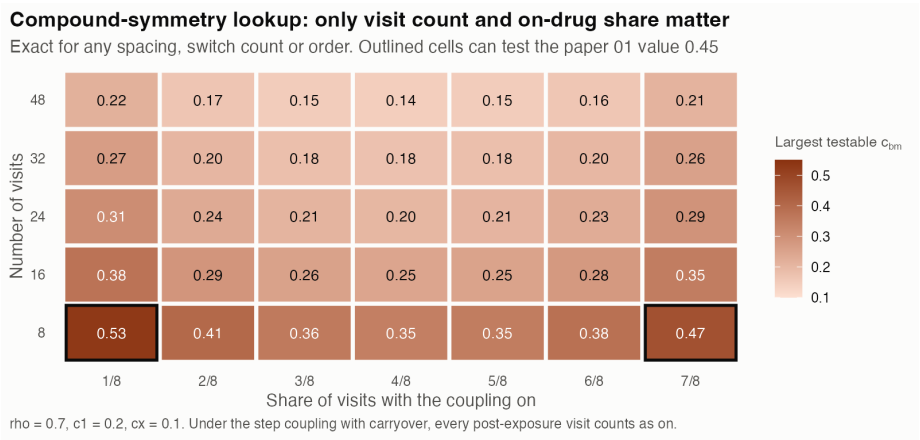


Figure 4: Largest testable c_{bm} under compound symmetry, from the closed form. Exact for any spacing, switch count or order. Under the step coupling with carryover, every post-exposure visit counts as on, so n_{on} is the number of visits from first exposure onward; $\rho = 0.7$, $c_1 = 0.2$, $c_\times = 0.1$.

Table 9: CS and AR(1) limits across cross-factor parameters c_1 and $c_\times$

c_1	$c_\times$	$c_1 - c_\times$	CS limit	AR(1) limit
0.1	0.0	0.1	0.346	0.485
0.2	0.1	0.1	0.346	0.465
0.3	0.2	0.1	0.345	0.435
0.1	0.1	0.0	0.378	0.504
0.2	0.2	0.0	0.377	0.491
0.2	0.0	0.2	0.261	not PD
0.3	0.1	0.2	0.261	not PD

Under AR(1) the difference also dominates, but the levels themselves matter too, and a large difference makes the response block itself invalid.

5.2 AR(1): schedule design governs the limit

Under AR(1) every schedule feature examined moves the limit (verified).

Table 10: Effect of schedule design decisions on the AR(1) ceiling

Design decision	Effect on the largest testable c_{bm} under AR(1)	Figure
More visits	Lowers it: 0.605 to 0.434 from 4 to 32 visits at 1-week spacing; 0.758 to 0.358 at 2.5-week spacing	5
More on/off switches	Lowers it sharply: 0.508 to 0.149 from 1 to 15 switches at 1-week spacing; 0.480 to 0.278 at 2.5-week spacing	6
Shorter gap at the switch	Lowers it: 0.723 at an 8-week gap to 0.331 at a half-week gap, at fixed study length	7
Uniform spacing	Non-monotone: 0.564 at 1 week, 0.619 at 2.5, 0.494 at 10; at 0.5 weeks the response block is not positive definite	7
Clustering on-drug visits	Raises it: 0.735 with 1-week gaps on drug and 4-week gaps off, against 0.555 for the reverse	3
More on-drug visits	Lowers it steadily: 0.867 with 1 of 8 on drug to 0.506 with 7 of 8	8
Order of on-drug visits	Matters: 0.526, 0.497 and 0.447 for three orders with the same visit count and on-drug count	3

Two features of these results bear directly on design choices. First, the number of switches is the most consequential decision under AR(1), and its cost depends on spacing: at 1-week spacing, going from one switch to three costs 0.15, whereas at 2.5-week spacing it costs 0.05. Multi-cycle N-of-1 designs, which exist to create many switches, are therefore the designs in which a covariance-moderation simulation can test the smallest biomarker correlations. Second, the gap across a switch matters more than the average spacing: holding study

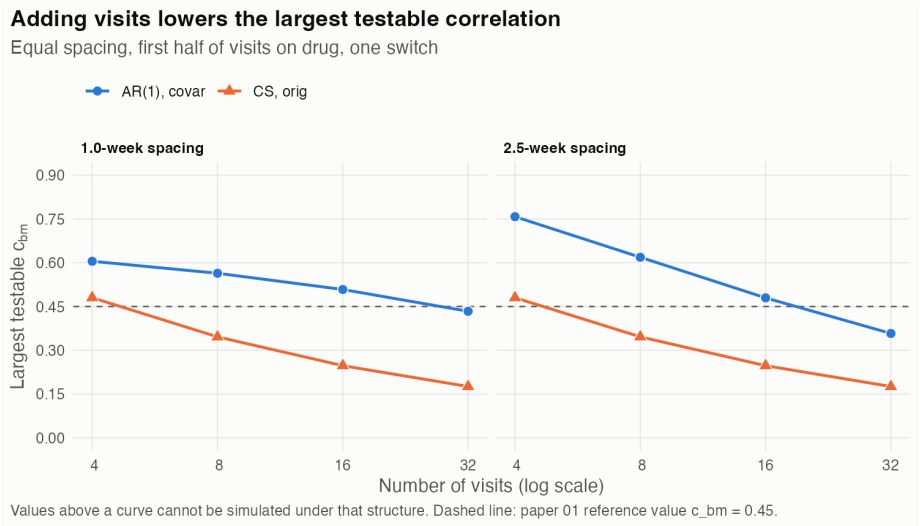


Figure 5: Largest testable c_{bm} by number of visits, at two uniform spacings, with the first half of visits on drug and one switch.

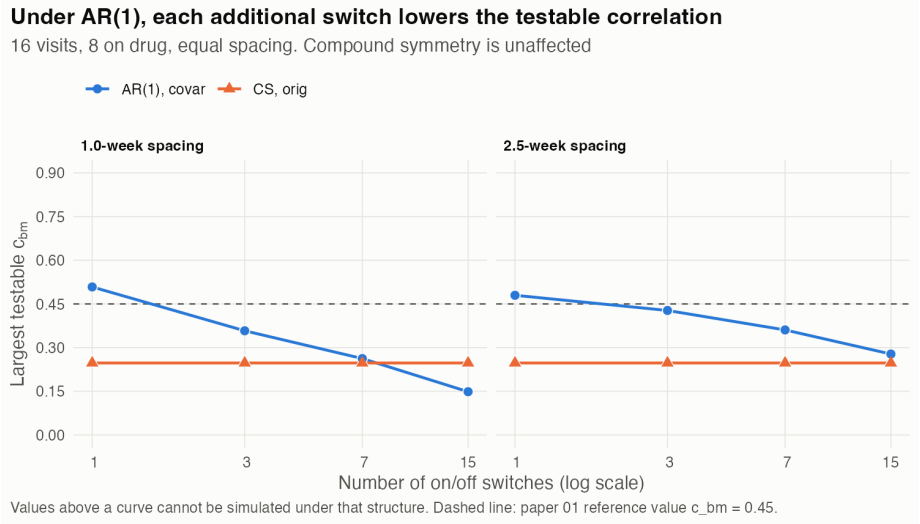


Figure 6: Largest testable c_{bm} by number of on/off switches, for 16 visits of which 8 are on drug. Under AR(1) each switch lowers the limit, and the cost per switch is larger at 1-week than at 2.5-week spacing. The compound-symmetry limit is unaffected.

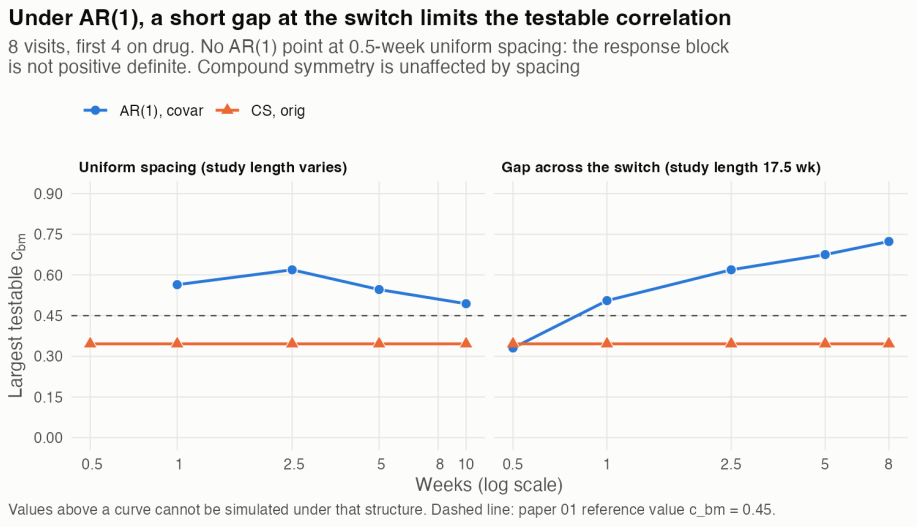


Figure 7: Largest testable c_{bm} by uniform spacing (left), where study length grows with spacing, and by the gap across the single switch at a fixed study length of 17.5 weeks (right).

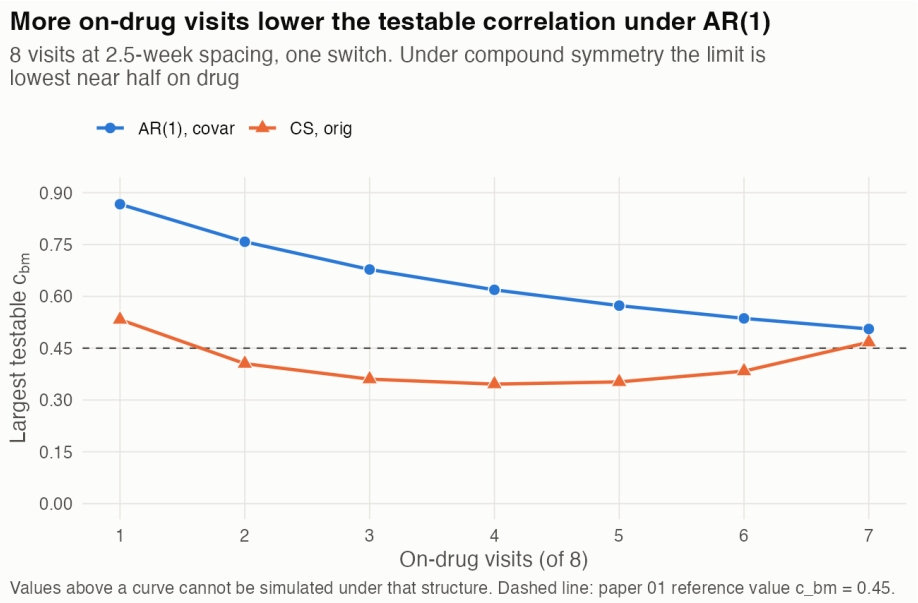


Figure 8: Largest testable c_{bm} by the number of on-drug visits, for 8 visits at 2.5-week spacing with one switch.

length fixed, widening only the switch gap from 1 to 2.5 weeks raises the limit from 0.505 to 0.619.

5.3 A lookup for spacing and switches

Figure 9 crosses the two most consequential decisions, visit spacing and number of switches, for a 16-visit design with half the visits on drug. It shows a trade-off that neither factor reveals alone. With a single switch, closer spacing gives the higher limit (0.52 at 1.5 weeks against 0.40 at 5 weeks), because the one switch is the only place where closeness is costly and a shorter study keeps the on-drug visits correlated. With many switches the ordering reverses (0.15 at 1 week against 0.33 at 5 weeks with fifteen switches), because every switch then pays the cost of a short gap. Only 4 of the 24 designs in the grid can test $c_{bm} = 0.45$, all with a single switch and spacing of 2.5 weeks or less.

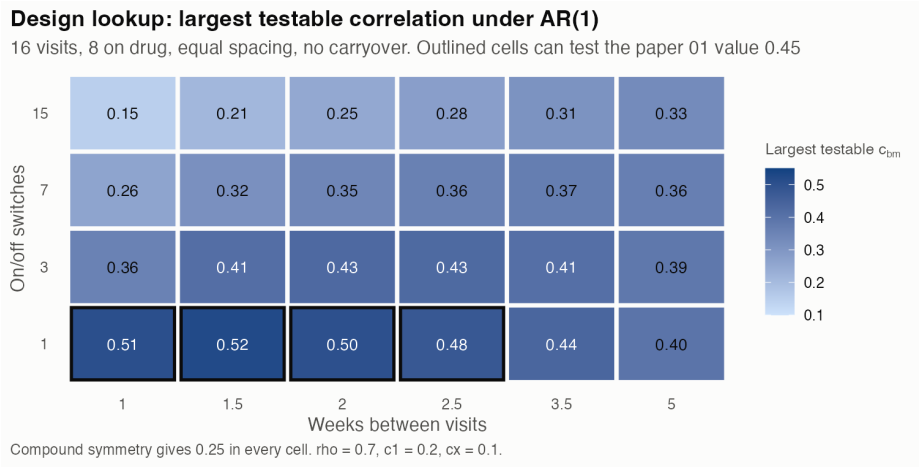


Figure 9: Largest testable c_{bm} under AR(1) for 16 visits, 8 on drug, equal spacing and no carryover. Outlined cells can test the paper 01 reference value 0.45. Compound symmetry gives 0.25 in every cell.

5.4 The paper 01 schedules

The Hybrid and OL+BDC schedules place one-week gaps at their switches. Replacing a path's visit times with evenly spaced times over the same study length, keeping the order of on-drug and off-drug visits, raises the AR(1) limit from 0.465 to 0.507 for Hybrid path C and from 0.452 to 0.545 for OL+BDC path A. (OL+BDC path A is that design's binding path. Hybrid path C is not: Hybrid's binding path at $t_{1/2} = 0$ is path A, at 0.456, and the equalized version of path A was not computed. Path C has four of eight visits on drug, as

in the synthetic schedules of Section 5.1. Correction of an earlier wording that called both the binding path, 2026-10-07, verified against `path.csv`.) The CO schedule is already evenly spaced. Uneven spacing therefore accounts for part of the gap between the CO limit (0.619) and the other two: just over half of it for OL+BDC (0.093 of 0.167), and about a quarter for Hybrid (0.042 of 0.154). The remainder for Hybrid is consistent with its three switches against CO's one (Section 5.2), although this attribution was not isolated directly.

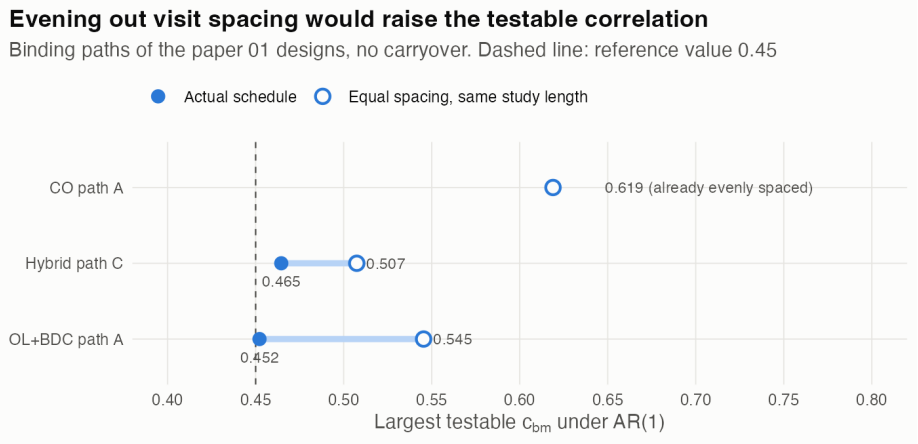


Figure 10: Largest testable c_{bm} under AR(1) for CO path A, Hybrid path C and OL+BDC path A, as scheduled and with evenly spaced visits over the same study length. The figure's subtitle calls these the binding paths; that holds for CO and OL+BDC but not for Hybrid, whose binding path A has 0.456 (Section 5.4).

5.5 Why AR(1) is sensitive to schedule design

For unit-variance variables, the biomarker's correlations with two response visits cannot differ by more than their own dissimilarity allows:

$$\left|\mathrm{Cor}(B, BR_t) - \mathrm{Cor}(B, BR_s)\right| \le \sqrt{2(1 - \mathrm{Cor}(BR_t, BR_s))}.$$

Covariance moderation requires exactly such a difference, from c_{bm} at an on-drug visit to about zero at the adjacent off-drug visit. Under AR(1), visits one week apart correlate at $\rho = 0.7$, so a switch across a short gap uses much of the available room, and each further switch uses more. Under CS every pair of visits correlates at ρ regardless of separation, so where a switch falls cannot matter.

The standard precision form of an AR(1) process with unequal gaps d_t makes the accounting explicit for the response block alone:

$$u^\top A^{-1} u = u_1^2 + \sum_{t \geq 2} \frac{(u_t - \rho^{d_t} u_{t-1})^2}{1 - \rho^{2 d_t}}.$$

A switch across a gap d contributes about $1/(1-\rho^{2d})$, which is large when d is short. A step between consecutive on-drug visits contributes $(1-\rho^d)/(1+\rho^d)$, which is small when d is short. Larger contributions mean a lower limit, which accounts for the direction of every effect in Section 5.2. The expression omits the cross-factor conditioning, so it explains the direction of the effects; the numbers in this section come from the full matrix.

5.6 Shape of carryover decay

The analyses above decay the biomarker-response correlation exponentially, as `buildSigma()` does. Paper 02 examines alternative decay shapes for the carryover mean, using the Weibull family $\phi(t) = \exp\{-(\lambda_w t)^k\}$ with $\lambda_w = (\ln 2)^{1/k}/t_{1/2}$, which passes through $\phi = 0.5$ at $t_{1/2}$ for every k (verified) and reduces to the exponential at $k = 1$. Shapes with $k < 1$ fall faster before the half-life and more slowly after it, giving a heavier tail; shapes with $k > 1$ give accelerated washout. We asked what the ceiling would be if the correlation decayed as each of the paper 02 shapes, $k \in \{0.25, 0.5, 1, 2, 4\}$.

Decay shape enters the ceiling only through the off-drug entries of the coupling vector. It does not affect M, and it has no effect at $t_{1/2} = 0$. Two consequences follow before any computation. First, because the binding cell for the paper 01 reference value is the no-carryover cell, decay shape cannot change whether $c_{bm} = 0.45$ is feasible in those designs. Second, changing only the shape of the carryover *mean*, as paper 02 does, leaves the ceiling unchanged, because the means do not enter the correlation matrix.

The computation added a fourth design to the paper 01 three: a multi-cycle schedule of 16 weekly visits in the repeating pattern on, on, off, off (seven switches), representing the many-switch designs of Section 5.2. Under Architecture B (verified):

Table 11: Architecture B ceilings by Weibull decay shape, design and half-life

Design	$t_{1/2}$	$k =$ 0.25	0.5	1 (exp.)	2	4	No carryover
Hybrid	0.5	0.543	0.531	0.506	0.469	0.456	0.456
Hybrid	1	0.551	0.551	0.544	0.527	0.518	
Hybrid	2	0.556	0.563	0.563	0.551	0.529	
OL+BDC	0.5	0.524	0.518	0.500	0.465	0.452	0.452
OL+BDC	1	0.530	0.533	0.533	0.523	0.516	
OL+BDC	2	0.534	0.540	0.546	0.542	0.526	

Design	$t_{1/2}$	$k =$ 0.25	0.5	1 (exp.)	2	4	No carryover
Multi-cycle	0.5	0.390	0.353	0.301	0.271	0.262	0.262
Multi-cycle	1	0.415	0.400	0.364	0.308	0.292	
Multi-cycle	2	0.440	0.448	0.451	0.430	0.395	
CO	any	0.619	0.619	0.619	0.619	0.619	0.619

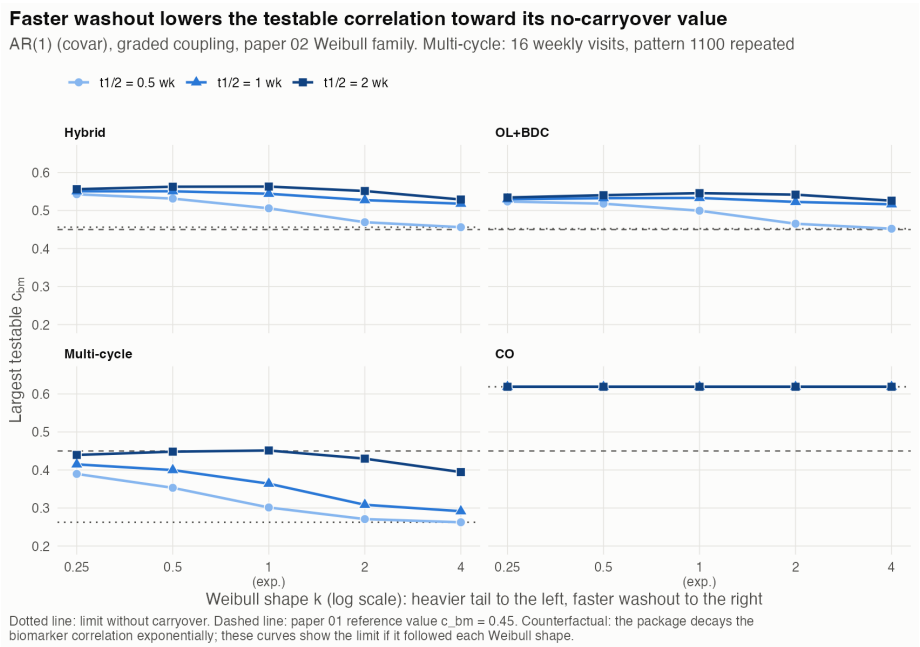


Figure 11: Largest testable c_{bm} under Architecture B if the biomarker-response correlation decayed as a Weibull curve of shape k, for three carryover half-lives. The dotted line is each design's limit without carryover. In CO the three half-life curves coincide.

Three patterns stand out.

- **At short half-lives, faster washout acts like less carryover.** At $t_{1/2} = 0.5$ and 1 the ceiling falls monotonically as k rises. At $k = 4$ and $t_{1/2} = 0.5$ the residual fraction one week after discontinuation is 1.5×10^{-5}, so every off-drug entry is effectively zero and Hybrid, OL+BDC and the multi-cycle design return exactly to their no-carryover limits. A heavier

tail keeps more residual at the sampled visits and raises the ceiling. The spread across shapes is 0.07 to 0.09 in Hybrid and OL+BDC and up to 0.13 in the multi-cycle design.

- **At long half-lives, the ordering is not monotone.** At $t_{1/2} = 2$ the ceiling peaks near $k = 0.5$ to 1, and $k = 0.25$ is lower. The off-drug visits in these designs fall one to three weeks after discontinuation, before or near a two-week half-life, where a small-k curve lies *below* the exponential. What governs the ceiling is the residual fraction at the visits actually sampled, not the heaviness of the tail as such.
- **CO is unaffected by shape.** With carryover present, its binding path is the placebo-first path, which carries no carryover.

Step coupling and floating-point underflow. Under the published step coupling, only whether the residual is nonzero matters. A Weibull curve never reaches zero analytically, so for $k \leq 2$ the step coupling is constant after exposure and the ceiling is 0.84 in Hybrid, OL+BDC and the multi-cycle design, as under the exponential. At $k = 4$ it is not: the residual underflows to exactly zero in double precision (8.6×10^{-78} at two weeks and 0 at four weeks for $t_{1/2} = 0.5$), the step coupling detects the zero, and the ceiling falls to 0.384 (Hybrid) and 0.467 (OL+BDC) at $t_{1/2} = 0.5$, and to 0.467 (Hybrid) at $t_{1/2} = 1$ (verified). The ceiling of a step-coupled process under fast washout is therefore determined by floating-point underflow rather than by the model, which is a further reason to regard a step on "any residual effect" as fragile.

5.7 Using the figures for design decisions

- **Identify the structure first.** Under CS (Figures 3 and 4), only the number of visits and the share with the coupling on can change the limit; no scheduling decision helps. Under AR(1) (Figures 5 to 10), the schedule is the principal lever.
- **Under AR(1), the decisions that raise the limit** are, in order of effect in these experiments: fewer on/off switches; a wider gap at each switch; fewer visits overall; clustering on-drug visits closely; and fewer on-drug visits.
- **Compute the limit for the actual schedule before simulating.** The synthetic results indicate direction and rough size, but the interaction between spacing and switch count (Figure 9) means the limit for a specific schedule should be computed with `analysis/scripts/quick-sim/cbm-ceiling/01-cbm-ceiling.R` or the closed form of Section 3.2.
- **When the decay shape is uncertain, plan against the no-carryover limit.** Fast washout returns the Architecture B ceiling to its no-carryover value (Section 5.6), which is the lowest the ceiling takes across the shapes and half-lives examined. A target c_{bm} below that value is feasible whatever the shape.

- **Leave a margin below the limit.** Because the positive-definiteness repair acts silently (Section 4.7), a target c_{bm} just above the limit would be simulated with altered correlations rather than rejected.
- **The limit is a feasibility bound, not a power statement.** A design that can test a larger c_{bm} is not thereby more powerful; the limit only bounds which values of the moderation parameter a covariance-moderation simulation can represent.

6. Implications

6.1 For paper 01

Note added 2026-10-07. Paper 01 has been restructured since these recommendations were written. The Appendix B.7 placeholder and Section 2.2.4 no longer exist; the current manuscript reports the ceilings in its section "Room for larger effect sizes" at both $\rho = 0.8$ (0.256 under compound symmetry) and $\rho = 0.7$ (0.346), with the separable AR(1) form at 0.440 and 0.474 respectively, and its Appendix B.2 replaces the silent repair by a stop. The bullets below are kept as written for the record.

- **Replace the Appendix B.7 placeholder.** The `[TO RE-VERIFY]` note asks for design-specific ceilings under the published specification and under Architecture B. Section 4.1 supplies them, subject to the caveat in Section 2 about the published code. Draft replacement text is given in Section 7.
- **Qualify the reversal statement.** Appendix B.7 should state that the reversal requires every path to begin on drug and depends on the compound-symmetric block (Section 4.6).
- **Qualify the claim that CS restricts c_{bm} more tightly.** Section 2.2.4 (the paragraph beginning "The compound-symmetry-to-AR(1) change") and Appendix B.2 state this as a general property. It holds at $\rho = 0.7$ and fails at $\rho = 0.3$. The explanation offered in Section 2.2.4, that "a constant correlation applied uniformly across many timepoints depletes the matrix faster than a correlation that decays with lag", is incomplete rather than wrong. The smallest eigenvalue of M under CS, $(1-\rho)-(c_1-c_\times)$, falls with ρ and does not depend on the number of occasions, so the response block itself is not depleted by added occasions. The ceiling, however, depends on $v^\top M^{-1} v$, and under CS it does fall with the number of occasions at a fixed on-drug share, roughly as $n^{-1/2}$ (Section 5.1, Figure 4). The CS ceilings are nearly identical across the three designs (0.346 to 0.352 at $t_{1/2} = 0$) because all three have eight occasions with four to six coupled. *(Corrected 2026-10-07: an earlier version attributed the CS ceiling to the conditioning of M alone, which contradicts Section 5.1.)*
- **State the margin at the reference value.** $c_{bm} = 0.45$ lies within 0.006 of the ceiling in the Hybrid and OL+BDC no-carryover cells. The

manuscript should report this, since any sensitivity analysis at a larger c_{bm} in those cells would engage the silent repair.
- **Constrain the Hendrickson comparison arm.** Under the published specification the ceiling at $t_{1/2} = 0$ is 0.346 to 0.352. The planned re-run against `58b32a9` cannot use $c_{bm} = 0.45$ in the no-carryover cells without the repair altering the published correlation structure. The comparison should either run at a common c_{bm} below 0.346 or report that the published specification is infeasible at the paper's reference value. At the published $\rho = 0.8$ the bound is tighter still, 0.256 to 0.262, below the published $c_{bm} = 0.3$ (added 2026-10-07; `docs/36`).

6.2 For the project

- **Correct the `CLAUDE.md` figures.** The project `CLAUDE.md` states that AR(1) expands the feasible range "from max 0.25 under CS to 0.49+ under AR(1)". Neither figure describes the published or the implemented configuration at production parameters. The ceiling is 0.346 to 0.352 for the published specification and 0.452 to 0.456 for Architecture B at $t_{1/2} = 0$. The value 0.25 matches CS with a decaying cross-factor block and graded coupling in Hybrid (0.247), a counterfactual combination (inferred). The value 0.49 is reached under Architecture B only once carryover is present. *Correction (2026-10-07):* the inference about 0.25 was mistaken. The figure matches the published configuration at its own $\rho = 0.8$, where the ceiling is 0.256 (Hybrid and CO) to 0.262 (OL+BDC) without carryover (`docs/36`, Section 4.2; verified). The statements above about 0.346 to 0.352 hold at $\rho = 0.7$ only. Under Architecture B at $\rho = 0.8$ the no-carryover ceiling is 0.291 (Hybrid) to 0.303 (OL+BDC) (verified).
- **Make the repair visible.** `buildSigma()` should either warn unconditionally when it repairs a matrix, or the drivers should call `validateParameterGrid()` before simulation and stop on failure. In either case the ceiling in Section 3.2 is cheap enough to compute for every cell of a grid before any simulation is run.
- **Decide whether the correlation should follow the mean's decay shape.** `buildSigma()` decays the biomarker-response correlation exponentially regardless of the shape used for the carryover mean. A pipeline that gives the mean a Weibull shape therefore decays the mean and the correlation with different shapes. Whether that is intended should be settled before decay-shape results are interpreted as properties of Architecture B.
- **Two defects in `nof1power::carryover_decay()`**, found while preparing Section 5.6 (both verified by running the function). The power form returns `NaN` for every time beyond three half-lives, because it raises a negative base to a fractional power before clamping at zero. And its documentation states that every form equals 0.5 at the half-life, whereas the power form with its default exponent 1.5 gives 0.544 there. Neither affects

this whitepaper, which uses only the Weibull family, but both affect any analysis that uses the power form.

7. Draft replacement for the Appendix B.7 placeholder

The following text is proposed in place of the `[TO RE-VERIFY]` paragraph at the end of the feasible-range discussion in Appendix B.7. It has not been inserted into the manuscript.

Note added 2026-10-07. Superseded. Paper 01 no longer has this appendix; its current ceiling table reports the published structure at $\rho = 0.8$ (0.256) and $\rho = 0.7$ (0.346), and the separable AR(1) form (0.440 and 0.474). The draft below states the published specification at $\rho_c = 0.7$ only and should not be quoted as the published code's ceiling.

> Evaluated on the three designs of Section 2.3 at the production parameters ($\rho_c = 0.7$ for all factors, $c_1 = 0.2$, $c_\times = 0.1$), the ceiling under Architecture B is 0.619 in CO, 0.456 in Hybrid and 0.452 in OL+BDC at $t_{1/2} = 0$, rising to 0.544 and 0.533 in Hybrid and OL+BDC at $t_{1/2} = 1.0$ and remaining at 0.619 in CO. The reference value $c_{bm} = 0.45$ is therefore feasible in every cell, but lies within 0.006 of the ceiling in the no-carryover cells of the Hybrid and OL+BDC designs. Under the published specification the ceiling is 0.346 to 0.352 at $t_{1/2} = 0$ in all three designs, and 0.841 at positive half-lives in Hybrid and OL+BDC, where every path begins on drug; in CO it remains 0.346. The published specification cannot carry $c_{bm} = 0.45$ without carryover. The comparison of within-factor structures depends on ρ_c: at $\rho_c = 0.7$ the AR(1) form admits the higher ceiling in every design, whereas at $\rho_c = 0.3$ compound symmetry does.

8. Limitations and what was not done

- **Published code.** The CS and step-coupling configurations were built from the manuscript's description of commit `58b32a9`, not from that code (Section 2). *Update 2026-10-07:* the specification has since been confirmed against the code (Section 2), but the published parameters use $\rho = 0.8$; see the review note at the top.
- **Parameter scope.** Ceilings were computed at $c_1 = 0.2$ and $c_\times = 0.1$ only, and for a common ρ across the three response factors. Unequal ρ_c values would reintroduce the loop-order dependence of the cross-factor decay rate in the current implementation.
- **Number of occasions.** Appendix B.7's statement that both ceilings fall in similar proportion as occasions are added was not tested. *Update*

2026-10-07: the schedule experiment of Section 5.2 (`determinants.csv`, experiment E1) bears on it. From 4 to 32 visits with half on drug, the CS ceiling falls from 0.480 to 0.176 (by 63%), the AR(1) ceiling from 0.605 to 0.434 at 1-week spacing (28%) and from 0.758 to 0.358 at 2.5-week spacing (53%) (verified). The two do not fall in similar proportion; CS falls faster.

- **Mean moderation.** Architecture A writes no biomarker-response correlation into R, so its covariance matrix has no ceiling of this kind, and it was not analyzed.
- **Power.** The ceiling bounds which values of c_{bm} can be simulated; it says nothing about power near the boundary. Whether near-singular matrices at $c_{bm} = 0.45$ affect the Monte Carlo behavior of the no-carryover cells was not examined.
- **Sensitivity sweep.** The ρ sweep covered Architecture B and CS with a constant cross-factor block only, not the two counterfactual combinations.
- **Design experiments without carryover.** The schedule experiments of Section 5 were run at $t_{1/2} = 0$ only. Carryover raises the Architecture B ceiling on the paper 01 designs (Section 4.5), but its interaction with switch count and spacing was not examined.
- **Synthetic schedules.** Section 5 varies one feature at a time on synthetic schedules. The attribution in Section 5.4 of the remaining Hybrid gap to its switch count is inferred, not isolated. The multi-cycle N-of-1 designs of papers 05 and 12 were not evaluated directly.
- **Decay shape is a counterfactual.** Section 5.6 computes the ceiling that would hold if the correlation decayed as a Weibull curve. The package does not implement that, and no simulation was run under it. The shapes examined are the Weibull family of paper 02 only.
- **Two-component DGPs.** The `components` argument permits a response built from BR and one other factor (papers 13 and 14). The ceilings here assume all three components; with two, the constants of the CS closed form change and the numbers would differ.

References

1. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Frontiers in Digital Health* 2020;2:13. doi:10.3389/fdgth.2020.00013. (Verified via Crossref, 2026-10-07.) Code: `github.com/rchendrickson/pmsimstats`, commit `58b32a9`.

9. Reproducibility

The analysis consists of four scripts with no Monte Carlo component. They are deterministic and together run in about a minute. Run them from the repository root, in order.

```
Rscript analysis/scripts/quick-sim/cbm-ceiling/01-cbm-ceiling.R
Rscript analysis/scripts/quick-sim/cbm-ceiling/02-ceiling-determinants.R
Rscript analysis/scripts/quick-sim/cbm-ceiling/03-decay-shape.R
Rscript analysis/scripts/quick-sim/cbm-ceiling/04-ceiling-figures.R
```

`01-cbm-ceiling.R` loads the package source with `pkgload::load_all('.')`, validates its matrix builder against `buildSigma()`, and checks the closed-form ceiling by bisection (Sections 3 and 4). `02-ceiling-determinants.R` runs the synthetic schedule experiments and checks the compound-symmetry closed form against the numerical ceiling (Sections 5.1 to 5.4). `03-decay-shape.R` computes the ceilings under the Weibull decay family (Section 5.6). `04-ceiling-figures.R` reads the saved tables and draws the eleven figures. Tables are written to `analysis/data/quick-sim/cbm-ceiling/` and figures to `docs/figures/`.

Table 12: Output files of the ceiling scripts and their contents

File	Contents
`design.csv`	Design-level ceilings (minimum over paths) and binding path, for all crossed configurations
`path.csv`	Path-level ceilings and smallest eigenvalue of M
`rho.csv`	Design-level ceilings across $\rho \in \{0.3, 0.5, 0.7, 0.9\}$
`determinants.csv`	Ceilings for every synthetic schedule of Section 5, both structures
`determinants-crossfactor.csv`	Ceilings across c_1 and $c_\times$ on Hybrid path C
`determinants-grid.csv`	Spacing by switch-count lookup grid (Figure 9)
`determinants-cs-grid.csv`	Compound-symmetry lookup by visit count and on-drug share (Figure 4)
`decay-shape.csv`	Design-level ceilings by Weibull shape, half-life and coupling (Figure 11)

The figures use categorical palette slots 1 and 2 of the reference design palette, validated for color-vision-deficiency separation and contrast on the light chart surface, with shape as a second encoding of structure. Figure 11 encodes the ordered half-life with a single-hue sequential ramp from the same palette, validated as an ordinal ramp, again with shape as a second encoding.